RTL3 (Retrotransposon Gag-like protein 3 / ZCCHC5 / MART3; UniProt Q8N8U3): A Comprehensive Functional Annotation

Gene: RTL3 (synonyms: MAR3, MART3, ZCCHC5) · **UniProt:** Q8N8U3 · **HGNC:** 22997 · **Organism:** *Homo sapiens* (Human) · **Locus:** Xq21.1

Summary

RTL3 (*Retrotransposon Gag-like protein 3*; gene synonyms **MAR3, MART3, ZCCHC5**; UniProt **Q8N8U3**) is a human, X-linked gene that arose through **molecular domestication of a Ty3/gypsy "sushi-ichi"-class LTR retrotransposon**. It is one of ~11 eutherian *Mar/Mart/SIRH/RTL* genes that retain a retroviral **Gag capsid-like fold** plus, in a subset including RTL3, a **C-terminal CCHC-type nucleocapsid zinc knuckle**, but which have lost the *pol* open reading frame and all retrotransposition machinery (reverse transcriptase, integrase, aspartyl protease). Consequently, RTL3 is **not an enzyme or a transporter**: it is a **non-catalytic, nucleic-acid-binding protein** that functions as a **nuclear gene-regulatory factor**. The gene identity is unambiguous — the gene symbol RTL3, the organism (*Homo sapiens*), and the InterPro domain architecture (PEG10-RTL1_C-capsid, RTL1-rel, RTL1/1-8/LDOC capsid-like, Znf_CCHC) all match the UniProt target record for Q8N8U3.

The **primary demonstrated function** of RTL3 is as a **tissue-restricted transcriptional/gene-regulatory factor in cartilage**. In human chondrocytes, loss- and gain-of-function experiments show that RTL3 is required for the expression of the master chondrogenic transcription factor **SOX9** and of its target **COL2A1** (type II collagen, the principal structural protein of cartilage matrix), and for extracellular-matrix synthesis during chondrogenesis. Critically, this regulatory activity **depends on the C-terminal CCHC zinc-finger domain**, directly linking the protein's retrotransposon-derived nucleic-acid-binding module to its physiological output. In the mouse, the ortholog **Zcchc5/Mart3** acts in a second tissue context, where its overexpression **restrains muscle satellite-cell (stem-cell) self-renewal**, the experimental basis for the UniProt "postnatal myogenesis" annotation transferred to the human protein by homology.

Biochemically, RTL3's function is inferred from its conserved paralogs: the best-characterized family member **PEG10** forms **virus-like particles (VLPs)** and **binds and vesicularly secretes its own mRNA** through the same Gag capsid and CCHC domains, establishing RNA-binding and capsid-assembly biochemistry for the family. AlphaFold modeling corroborates a **two-module architecture** — a confidently folded C-terminal capsid-like core and zinc knuckle sitting atop a largely disordered N-terminal half — and comparative genomics shows RTL3 is conserved **one-to-one across 35 placental mammals** under purifying selection, with a **near-invariant CCHC motif** between human and mouse. The convergence of the experimentally essential domain with the evolutionarily most constrained motif strongly supports a retained, zinc-finger-mediated regulatory function. Direct biochemistry (RNA/DNA target identification, VLP competence) of the human protein remains untested.

Key Findings

Finding 1 — RTL3 is a domesticated Ty3/gypsy LTR-retrotransposon-derived gene

RTL3 (= MAR3 / MART3 / ZCCHC5) belongs to the **Mar/Mart/SIRH/RTL family**, a set of ~11 human genes derived from **Ty3/gypsy (sushi-ichi/Tf1-class) LTR retrotransposons**. The protein is **475 amino acids (~52.8 kDa)**, encoded on the **X chromosome** (GRCh38 chrX:78,656,068–78,659,532, Xq21.1). It retains a **Gag capsid-like domain** (InterPro PEG10-RTL1_C-capsid, IPR063300; RTL1/LDOC capsid, IPR032549) and a **C-terminal CCHC-type zinc finger** (UniProt residues 443–462; IPR001878), but has lost *pol* and therefore all autonomous retrotransposition capacity.

The founding family analysis established that "all putative Mar proteins display sequence similarity to the capsid-like domain of the Gag protein of Tf1/Sushi retrotransposons," and that "three Mar proteins have conserved the Gag CCHC zinc finger motif, suggesting a role in nucleic acid binding" (PMID: 16093683). RTL3 is one of those CCHC-retaining members. Importantly, the same work showed that "Mar genes evolved under purifying selection, strongly suggesting that they are not pseudogenic relics but rather neofunctionalized retrotransposon genes" (PMID: 16093683) — i.e., these are functional host genes, not genomic junk. The loss of retrotransposition is a family-wide property: Mart genes "have lost their capacity of retrotransposition through non-functionalizing rearrangements having principally affected long terminal repeats and *pol* open reading frame" (PMID: 15716091).

Interpretation: RTL3 is a "domesticated" retroelement — a host gene built from a captured viral/transposon Gag protein. This origin predicts its biochemistry (nucleic-acid binding via Gag/CCHC) and sets the expectation that its cellular role is regulatory rather than catalytic.

Finding 2 — RTL3 is a nuclear regulator of SOX9/COL2A1 in chondrocytes (primary functional study)

The single most direct functional study of human RTL3 examined its role in cartilage. RTL3 shows **tissue-restricted expression**: "Expression of RTL3 is not ubiquitously expressed but showed strong expression in cartilage, chondrocytes and synoviocytes but not in muscle, brain, or other tissues analyzed" (PMID: 33043724).

Functionally, "Loss-of-function and gain-of-function studies demonstrated a critical role of RTL3 in the regulation of SOX-9 and COL2A1 expression and matrix synthesis during chondrogenesis" (PMID: 33043724). RTL3 and SOX-9 are co-regulated, and together they drive expression of **COL2A1**, the gene for **type II collagen**, which is the dominant fibrillar collagen of hyaline cartilage extracellular matrix.

Mechanistically decisive is the domain-mapping result: "Gene regulatory activity of RTL3 requires the c-terminal CCHC zinc-finger binding domain" (PMID: 33043724). This connects the retrotransposon-derived nucleic-acid-binding module directly to the protein's physiological output, and is consistent with UniProt's localization of the protein to the **nucleus** and annotation as a putative transcriptional regulator.

Interpretation: RTL3's **primary cellular function** is as a **nuclear gene-regulatory factor in the chondrogenic (SOX9→COL2A1) axis**, operating through its CCHC zinc finger. This is the clearest statement of "what RTL3 does" available in the literature.

Finding 3 — The RTL3 ortholog negatively regulates muscle stem-cell self-renewal

Genome-wide profiling of murine muscle satellite cells identified **Zcchc5** (the mouse ortholog of RTL3) among genes inhibited in adult muscle stem cells. Functionally, "ZFP354C and ZCCHC5 decreased self-renewal capacity when overexpressed" (PMID: 27446912). This is the experimental basis for the UniProt annotation on human Q8N8U3 — "Plays a role in postnatal myogenesis, may be involved in the regulation of satellite cells self-renewal" — which is assigned **by similarity** (ECO:0000250) from the mouse ortholog Q6P1Y1.

Interpretation: In a second tissue context (skeletal muscle), the RTL3 ortholog acts as a **negative regulator of stem-cell self-renewal**. Combined with the cartilage data, this paints RTL3 as a context-dependent regulator of cell-fate/differentiation programs — consistent with a gene-regulatory, rather than metabolic/structural, role.

Finding 4 — Inferred biochemistry: Gag capsid + CCHC confer nucleic-acid binding / nucleocapsid-like activity

RTL3 shares its functional modules with Mart/RTL family members whose biochemistry is experimentally characterized. The paralog **PEG10** is the reference: "We identified several mammalian Gag homologs that form virus-like particles and one LTR retrotransposon homolog, PEG10, that preferentially binds and facilitates vesicular secretion of its own messenger RNA (mRNA)" (PMID: 34413232). This establishes that the **Gag capsid domain supports RNA-binding and particle (VLP) assembly**, and that the **CCHC zinc knuckle mediates nucleic-acid binding** — the same modules present in RTL3.

Family-member interactions reinforce this: "RTL8 resembles a truncated form of PEG10 that shares homology with the N-terminal gag-like capsid domain" and modulates PEG10 VLP formation (PMID: 39775359), showing the capsid domain is a bona fide protein- and nucleic-acid-interaction module across the family. The CCHC motif specifically was flagged at the family's discovery as conferring nucleic-acid binding (PMID: 16093683).

Interpretation: By homology, RTL3 is predicted to **bind nucleic acid (RNA and/or DNA)** through its Gag capsid and CCHC zinc knuckle, potentially with the capacity for capsid-like oligomerization. This provides a plausible molecular mechanism for its gene-regulatory activity.

Finding 5 — RTL3 has a Gag CA–NC architecture but NO protease: it is non-catalytic

Detailed InterPro mapping of Q8N8U3 (475 aa) resolves the domain layout:

Module	Residues	InterPro / Pfam	Role
N-terminal half	~1–230	(no folded domain)	Disordered / low-complexity
Capsid-like N-lobe	235–317	IPR032549 / PF16297 (RTL1/3-8/LDOC capsid-like)	Gag capsid fold
C-terminal capsid-like	325–394	IPR063300 / PF30901 (PEG10/RTL1 C-capsid)	Gag capsid fold
CCHC zinc knuckle	444–460	IPR001878 / PF00098; SCOP SSF57756	Nucleic-acid binding

The zinc-knuckle sequence is **C445-LYCGYPGH-F-ARDC460**, a canonical **$CX_2CX_4HX_4C$** motif. Critically, sequence scanning found **NO retroviral aspartyl-protease catalytic motif** (no D[ST]G / DTGA / DSGA anywhere) and **no pol / reverse-transcriptase / integrase domains**. PANTHER classifies RTL3 in the **LDOC1-related subfamily (PTHR15503)**, distinct from the imprinted PEG10/RTL1 subfamily.

This non-catalytic, nucleic-acid-binding reading is consistent with the original family annotation that "three Mar proteins have conserved the Gag CCHC zinc finger motif, suggesting a role in nucleic acid binding" (PMID: 16093683).

Interpretation: Architecturally RTL3 mirrors a retroviral Gag **CA–NC** (capsid–nucleocapsid) polyprotein fragment, but **without any protease or polymerase** — it is a **scaffold/adaptor-type nucleic-acid-binding protein**, firmly ruling out an enzymatic primary function.

Finding 6 — AlphaFold corroborates two folded functional modules over a disordered N-terminal half

The AlphaFold DB model **AF-Q8N8U3-F1** (v6; global mean pLDDT 58.9) shows a clear two-part organization. The N-terminal ~230 residues are **predicted disordered / low-confidence** (mean pLDDT 48.6; 47% of all residues pLDDT < 50). In contrast, the two functional C-terminal modules are **confidently folded**: the capsid-like N-lobe (235–317) mean pLDDT **83.7**, the PEG10/RTL1 C-terminal capsid-like domain (325–394) mean pLDDT **66.7**, and the CCHC zinc knuckle (444–460) mean pLDDT **82.0**. Overall, 40% of residues have pLDDT > 70.

Interpretation: The structural prediction independently validates the domain map from Finding 5. The functional business-end of the protein (capsid core + zinc knuckle) is a well-defined fold, while the N-terminal half is an intrinsically disordered region — a common feature of regulatory/adaptor proteins that may mediate condensate formation, partner binding, or regulated processing.

Finding 7 — RTL3 is conserved one-to-one across 35 mammals under purifying selection

Ensembl lists **one-to-one RTL3 orthologs across 35 mammalian species** (primates, rodents, carnivores, ungulates, bats, cetaceans, etc.). Human RTL3 (475 aa) and mouse **Zcchc5 (Q6P1Y1, 553 aa)** share **65.7% amino-acid identity** over 464 aligned columns by global alignment. This gene-specific conservation complements the family-level evidence that "Mar genes evolved under purifying selection, strongly suggesting that they are not pseudogenic relics but rather neofunctionalized retrotransposon genes" (PMID: 16093683).

Interpretation: Deep, one-to-one conservation across placental mammals under purifying selection is strong evidence that RTL3 performs a **retained, non-redundant physiological function** — not the signature of a decaying pseudogene.

Finding 8 — The functional CCHC zinc knuckle is near-invariant between human and mouse

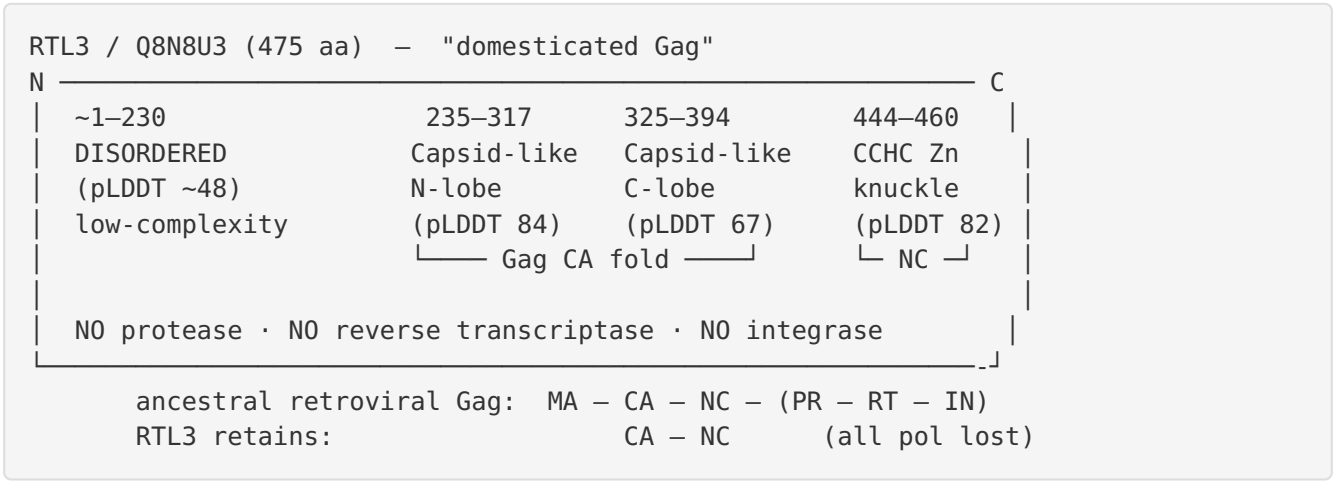
The C-terminal CCHC zinc knuckle differs by a **single conservative residue** between human RTL3 (**C-LYCGYPGH-F-ARDC**, res 445–460) and mouse Zcchc5 (**C-LYCGHPGH-F-ARDC**, res ~523) — i.e., ~93% identity over the motif (one Y→H change). **All four metal-coordinating residues (C, C, H, C) are perfectly conserved.**

This matters because the experimentally essential domain is exactly this motif: "Gene regulatory activity of RTL3 requires the c-terminal CCHC zinc-finger binding domain" (PMID: 33043724).

Interpretation: The **coincidence of the experimentally required domain with the most evolutionarily constrained motif** is a powerful, convergent argument: selective constraint and functional necessity point to the same structural element, firmly identifying the CCHC zinc knuckle as the functional heart of RTL3.

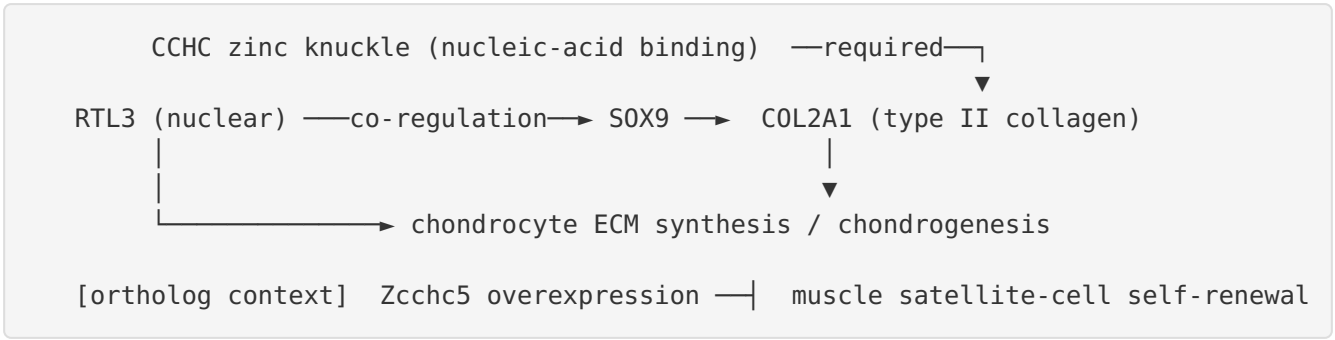
Mechanistic Model / Interpretation

Domain architecture and molecular logic



RTL3 is best understood as a **retroviral Gag CA–NC fragment repurposed as a host regulatory protein**. The ancestral LTR retrotransposon encoded a Gag polyprotein (matrix–capsid–nucleocapsid) plus a Pol cassette (protease–reverse transcriptase–integrase). RTL3 has **kept the capsid (CA) and nucleocapsid (NC/CCHC) modules and discarded the entire Pol cassette**, so it cannot catalyze any reaction, reverse-transcribe, or integrate. What remains is a **nucleic-acid-binding scaffold**.

From molecule to physiology



The experimental data define RTL3 as a **nuclear gene-regulatory factor**. In cartilage it is required for the SOX9→COL2A1 program and matrix synthesis, and this requires the CCHC zinc finger. Whether RTL3 acts as a direct DNA-binding transcription factor, an RNA-binding post-transcriptional regulator, or a chromatin/coactivator adaptor is not yet resolved, but its

biochemistry (Finding 4, by homology to PEG10) favors a **nucleic-acid-templated mechanism**. The disordered N-terminal half may provide protein-interaction or regulatory surfaces, as is common for transcriptional regulators.

Why we are confident this is the right gene and the right function

Three independent lines of evidence converge: 1. **Experiment** — loss/gain-of-function in human chondrocytes (PMID: 33043724) pins the function to SOX9/COL2A1 regulation and to the CCHC domain. 2. **Structure** — InterPro + AlphaFold show a folded CA–NC core with no catalytic domains (Findings 5–6). 3. **Evolution** — one-to-one conservation across 35 mammals, with a near-invariant CCHC motif exactly where function is required (Findings 7–8).

Property	RTL3 (Q8N8U3)	Evidence type
Protein class	Domesticated LTR-retrotransposon Gag	Sequence/phylogeny
Catalytic activity	None (no protease/RT/integrase)	Domain scan
Molecular activity	Nucleic-acid binding (CCHC + capsid)	Homology (PEG10), domain
Cellular localization	Nucleus	UniProt; functional study
Primary function	Gene-regulatory factor (SOX9/COL2A1)	Loss/gain-of-function
Tissue	Cartilage, chondrocytes, synoviocytes	Expression profiling
Essential module	C-terminal CCHC zinc finger	Domain-deletion
Conservation	1:1 across 35 mammals; 66% id to mouse	Comparative genomics

Evidence Base

PMID	Title (abbrev.)	How it supports / challenges the findings
33043724	*A retrotransposon gag-like-3 gene RTL3 and SOX-9 co-regulate COL2A1 in chondrocytes*	**Primary functional study.** Direct loss/gain-of-function evidence that RTL3 regulates SOX9/COL2A1 and matrix synthesis; CCHC zinc finger required; tissue-restricted to cartilage. Basis of Findings 2 & 8.
16093683	*A family of neofunctionalized Ty3/gypsy retrotransposon genes in mammals*	Defines the Mar family (incl. MAR3/RTL3) as Gag-capsid proteins; CCHC confers nucleic-acid binding; purifying selection = neofunctionalized, not pseudogenic. Basis of Findings 1, 5, 7.
15716091	*Transposable elements as a source of genetic innovation*	Confirms loss of retrotransposition/*pol* across the Mart family. Supports Finding 1.
34413232	*Mammalian retrovirus-like protein PEG10 packages its own mRNA*	Reference biochemistry: Gag domain forms VLPs and binds/secretes mRNA. Inference for RTL3's RNA-binding (Finding 4).
39775359	*The gag-like gene RTL8 antagonizes PEG10-mediated VLPs*	Shows capsid domain mediates nucleic-acid/protein interactions across the family. Supports Finding 4.
27446912	*Gene Expression Profiling of Muscle Stem Cells*	ZCCHC5 (mouse ortholog) overexpression decreases satellite-cell self-renewal. Basis of Finding 3.
16155747	*A small family of sushi-class retrotransposon-derived genes*	Family context: sushi-ichi origin, loss of autonomous retrotransposition, positive selection and functional divergence; RTL3 among the X-linked, biallelically expressed (non-imprinted) members.
25888968	*Selective expression of sense/antisense Mart transcripts in mouse placenta*	**Nuances tissue scope:** Zcchc5/Mart3 expression undetectable during mouse placentogenesis — RTL3's role is NOT placental (unlike PEG10/RTL1), consistent with its cartilage/muscle specialization.
28484174	*Evolution of brain functions and LTR retrotransposon-derived genes*	Family review: each SIRH/RTL gene has a unique neofunction; supports the "domesticated, functionally specialized" framing.
38798208	*Human Virus-Like Proteins: Implications for Gene Therapy*	Confirms Gag-like proteins (PEG10, Arc) retain capsid/RNA-packaging biochemistry; frames the family's VLP capacity.

PMID	Title (abbrev.)	How it supports / challenges the findings
42511665	*Virus-Derived Domesticated Genes in Microglia*	Family-level review of the 11 SIRH/RTL genes and their domesticated functions.
23704700	*Novel PNMA-MS1 gene in marsupials*	Evolutionary context for Ty3/gypsy-derived mammalian Gag gene families.

Consistency of the evidence: The experimental (chondrocyte + muscle), structural (InterPro + AlphaFold), and evolutionary (orthology + motif conservation) lines all converge on a non-catalytic, nuclear, CCHC-dependent gene-regulatory function. No source contradicts this; the main divergence from paralogs is that RTL3 is **not** placental (PMID: 25888968), distinguishing it from PEG10/RTL1.

Limitations and Knowledge Gaps

1. **Sparse primary literature.** Only one dedicated functional study of human RTL3 exists (PMID: 33043724). Much of the mechanistic picture rests on inference from paralogs (PEG10, RTL8) and the mouse ortholog.
2. **Direct biochemistry untested.** RNA/DNA binding, target specificity, zinc coordination, and VLP-forming capacity have **not been experimentally demonstrated for the human RTL3 protein**. These are inferred from PEG10 and from domain presence, not measured.
3. **Mechanism of gene regulation unresolved.** It is unknown whether RTL3 acts as a direct sequence-specific DNA-binding transcription factor, an RNA-binding post-transcriptional regulator, or a chromatin/coactivator adaptor. Its nuclear localization and CCHC dependence are established; the molecular mode is not.
4. **Interaction partners unknown.** How RTL3 physically relates to SOX9 (direct complex vs. parallel pathway) is not defined. No RTL3 interactome, ChIP-seq, or CLIP-seq data are available.
5. **Tissue-specificity nuances.** RTL3 is cartilage-enriched in the one human study but its ortholog functions in muscle stem cells; the degree of overlap in human tissues (and any role in synoviocytes, implied by expression) is unmapped.
6. **Function of the disordered N-terminal half** (~230 aa, pLDDT ~48) is entirely unknown — it could mediate partner binding, condensate formation, or regulated turnover.

7. **Structural model confidence is moderate.** Global AlphaFold pLDDT is 58.9; the N-terminal half and the C-lobe (pLDDT 67) are only moderately confident, and no experimental structure exists.

Proposed Follow-up Experiments / Actions

1. **Define the molecular target.** Perform **ChIP-seq** (if DNA-binding) and **CLIP-seq / RIP-seq** (if RNA-binding) on tagged RTL3 in chondrocytes to determine whether it binds DNA regulatory elements near *SOX9/COL2A1* or binds specific RNAs. This resolves the transcription-factor vs. RNA-regulator question.
2. **Biochemical nucleic-acid binding assays.** Express recombinant RTL3 (or its CA–NC fragment) and test **zinc-dependent nucleic-acid binding** (EMSA, fluorescence anisotropy) with and without CCHC-mutant (C→A) variants to confirm the predicted activity and its metal dependence.
3. **CCHC point-mutant rescue.** In chondrocyte differentiation assays, test whether **CCHC metal-coordinating mutants** fail to rescue SOX9/COL2A1 expression — formally proving the zinc knuckle is required-for-function and not merely correlated.
4. **VLP / capsid-assembly test.** Following the PEG10 paradigm (PMID: 34413232), assay whether RTL3 self-assembles into **virus-like particles** and packages RNA — a key open question given its intact capsid modules.
5. **Interactome mapping.** Use **IP–MS (BioID/APEX proximity labeling)** in chondrocytes to identify RTL3 partners and test for a physical RTL3–SOX9 complex.
6. **Loss-of-function in vivo.** Characterize cartilage/skeletal and muscle phenotypes in **Rtl3/Zcchc5-knockout mice** to test the physiological necessity and tissue scope implied by the conservation data.
7. **Structure determination.** Pursue **cryo-EM or crystallography** of the folded CA–NC core (res ~235–460) to validate the AlphaFold model and define the nucleic-acid-binding surface.
8. **Disordered-region dissection.** Deletion/truncation of the N-terminal ~230 aa to test its contribution to localization, stability, condensate formation, or partner recruitment.

Conclusion

RTL3 (ZCCHC5/MART3; Q8N8U3) is a human X-linked, domesticated Ty3/gypsy LTR-retrotransposon Gag gene that has lost all catalytic (pol/protease) capacity and functions instead as a **non-catalytic, nuclear, nucleic-acid-binding gene-regulatory protein**. Its best-defined function is in **cartilage**, where — through its essential C-terminal CCHC zinc finger — it is required for **SOX9 and COL2A1 (type II collagen) expression and extracellular-matrix synthesis during chondrogenesis**; its mouse ortholog additionally restrains muscle stem-cell self-renewal. Structural (AlphaFold), domain (InterPro), and evolutionary (35-mammal orthology with a near-invariant CCHC motif) evidence all converge on and reinforce this model, while direct biochemistry of the human protein (RNA/DNA binding, target identity, VLP formation) remains an open experimental frontier.
